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Labour Mobility

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Notations

n.a.	:	not available
‘ - ’	:	nil or negligible
s	:	data is suppressed as the sample size is too small

List of Abbreviations

LFS	:	Labour Force Survey
MOM	:	Ministry of Manpower
OCL	:	Plant & Machine Operators & Assemblers, and Cleaners, Labourers & related workers
PMET	:	Professionals, Managers, Executives and Technicians
SCC	:	Sales & Service workers, Clerical workers, Production Craftsmen and related workers
SSOC	:	Singapore Standard Occupational Classification
WSQ	:	Workforce Skills Qualifications
WIS	:	Workfare Income Supplement

List of Occupation Abbreviations

Mngrs	:	Managers & Administrators
Prof.	:	Professionals
Tech.	:	Associate Professionals & Technicians
Clerk	:	Clerical Workers
Sales	:	Service & Sales Workers
Prod.	:	Production Craftsmen & Related Workers
Plant Op.	:	Plant & Machine Operators & Assemblers
Cleanr	:	Cleaners, Labourers & Related Workers

List of Industry Abbreviations

Manu.	:	Manufacturing
Const.	:	Construction
W.Sale & Retail	:	Wholesale & Retail Trade
Tpt & Str	:	Transport & Storage
Htl & Rst	:	Hotels & Restaurants
ICT	:	Information & Communication
Fin Serv	:	Financial Services
Real Estate	:	Real Estate & Leasing Activities
Prof Serv	:	Professional Services
Admin & Suppt	:	Administrative & Support Services
Comm & Social	:	Community, Social & Personal Services

HIGHLIGHTS

- In 2009, one in five (20%) resident employees changed jobs in the last two years. Among the resident employees who switched jobs, most moved to occupations with similar skills level. More employees also moved within industries than across industries.

- Workers who had undergone Workforce Skills Qualification (WSQ) training experienced greater job mobility. Close to three in ten (28%) employees with WSQ changed jobs in the last two years, compared with two in ten (19%) employees without WSQ training. Results of an econometric analysis, which isolated the impact of various factors, confirmed that WSQ training enhanced workers' mobility, significantly increasing the chances of plant & machine operators, cleaners and labourers moving to service, clerical, craftsmen & related occupations. WSQ also benefited professionals, managers, executives & technicians (PMETs) and those in service, clerical, craftsmen & related occupations, increasing their odds of transiting to similarly-skilled occupations.

- Reflecting the wide spectrum of WSQ programmes for services occupations, resident employees with WSQ were more likely to move within services after a job switch than those without WSQ. Similarly, WSQ training facilitated movement of workers from manufacturing to services.

- Labour mobility varied less significantly with education, with degree (21%) and diploma (20%) holders only slightly more likely to change jobs than those without secondary qualifications (19 to 20%). The econometric results showed that higher formal qualification, especially university education strongly facilitated the move to PMET occupations.

- Labour mobility generally decreased with age. Nearly three in ten (28%) workers aged below 30 changed jobs in the last two years. This decreased progressively to around one in ten (11%) for those aged above 60. The econometric analysis confirmed that older resident employees had a lower probability of changing jobs as individuals become more settled in their careers and shoulder increased responsibilities with age.

- The most mobile employees were from hotels & restaurants (27%), information & communications (25%), professional services (24%), administrative & support services (24%) and construction (24%). In contrast, employees from community, social & personal services were the least likely to have changed jobs (14%).

Labour Mobility

1 Introduction

Objective

1.1 Labour mobility[1] refers to the movement of workers across occupations (occupational mobility) and industries (industrial mobility). In particular, occupational mobility can be sub-categorised into horizontal (within a broad class of jobs similar in socio-economic status) or vertical (within a broad class of jobs non-similar in socio-economic status) movements.

1.2 At the aggregate level, the mobility of workers brings about important economic benefits by facilitating labour supply in new growth industries, value adding to existing sectors and adapting to structural and technological changes.[2] At the individual level, mobility can lead to improvements in the economic situation and/or sustain employment.

1.3 This study assesses the general determinants of labour mobility among resident[3] employees. It also examines the mobility of workers across various occupational skill-sets and industries. However, the workplace effects of mobility (e.g. working relationship with boss and work culture) are not explored here.

Data and Empirical Approach

1.4 The study makes use of individual-level anonymised data from the June 2009 Labour Force Survey (LFS).[4] The LFS collects data on the economic activities of the population, including detailed information on employment and unemployment, characteristics of the labour force and economically inactive persons. A total of 28,500 households responded to the June 2009 LFS, yielding data on close to 100,000 individuals with an overall response rate of 90.4%.

1.5 The study defines labour mobility as the proportion of currently employed residents engaged in salaried employment[5] who reported an occupation change[6] over the last two years (June 2007 – June 2009). The sample for this study consists of nearly 36,000 individual observations.[7]

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[1] Labour mobility also includes movement of workers across geographies (within or across economies), which is not the focus of this study. This study also does not distinguish between voluntary and involuntary mobility.

[2] Beyond economy-wide benefits, labour mobility also impacts social mobility and associated inequality issues at both the intra- and inter-generational levels. These are not covered in the current study as such analysis would require a longitudinal dataset.

[3] Residents refer to Singapore citizens and permanent residents.

[4] Ideally, having a sufficiently long and representative longitudinal data on individuals would enable researchers to control for a range of individual characteristics (observed and unobserved) and macro-economic dynamics like business cycles. As longitudinal data are not available, the study makes use of cross-sectional anonymised individual-level data from the LFS.

[5] They exclude the self-employed.

[6] A reported occupation change does not necessarily imply that the individual worker switched employers as the worker could have switched jobs within the company.

[7] In this study, it is assumed that the socio-demographic characteristics of an employee in June 2009 were the same two years ago (June 2007). For those who had changed jobs over the last two years, data of their previous occupation and industry were used in this study. .

Methodology

1.6 The study adopts a Discrete Limited Dependent Variable model to examine the impact of various factors on the labour mobility of the resident workforce, holding other factors constant. The probability of labour mobility is modelled as follows:

$$L = Log\left(\frac{P_i}{1 - P_i}\right) = \alpha + \beta_1 X_i + \beta_2 Z_i + \beta_3 Q_i$$

where P_i is the probability of labour mobility for the employed resident (individual i). X_i is a vector of personal characteristics including education, age, marital status, gender; Z_i is a vector of individual background, such as WSQ; Q_i is a vector of job-related characteristics, such as current labour income, current type of employment and current nature of employment.

1.7 Additionally, the study reports the odds ratio for easier interpretation. The odds ratio is the ratio of the probability of an event occurring in one group to the probability of it occurring in another group, in this case the reference group. If the probabilities of the event in each of the groups are P_1(first group) and P_2(reference group), then

$$odds\ ratio = \frac{P_1}{P_2}.$$

An odds ratio of 1 indicates that the condition or event under study is equally likely to occur in both groups; an odds ratio greater than 1 indicates that the condition or event is more likely to occur in the first group; and an odds ratio less than 1 indicates that the condition or event is less likely to occur in the first group.

2 Descriptive Statistics

General Trend

2.1 In 2009, one in five (20%) resident employees changed jobs in the last two years (Table 1). **Workers who had undergone Workforce Skills Qualification (WSQ) training experienced greater job mobility.** Close to three in ten (28%) employees with WSQ changed jobs in the last two years, compared with two in ten (19%) employees without WSQ training. Labour mobility varied less significantly with education, with degree (21%) and diploma (20%) holders only slightly more likely to change jobs than those without secondary qualifications (19 to 20%).

2.2 Labour mobility generally decreased with age. Nearly three in ten (28%) workers aged below 30 changed jobs in the last two years. This decreased progressively to around one in ten (11%) for those aged above 60. The differences in labour mobility among other demographic characteristics were less significant. Male employees (21%) were more likely to switch jobs than female employees (18%). Probably with less financial commitments, singles were more mobile (24%) than married (18%) or divorced / widowed (22%) employees.

2.3 Contract employees (34%) were nearly twice more likely to switch jobs than permanent (18%) employees. Similarly, "involuntary" part-timers (27%) (defined as those who were willing and available to work additional hours) were more mobile than full-timers (19%). On the other hand, "voluntary" part-timers were the least likely (17%) to change jobs.

2.4 Labour mobility decreased with income. As many as one in four (25%) employees earning a gross monthly income of less than $1,500[8] switched jobs over the past two years. This decreased to 20% for those earning $1,500 to less than $2,500[9], 18% of those earning $2,500 to below $4,333 and 15% for those earning above $4,333.

2.5 The most mobile employees were from hotels & restaurants (27%), information & communications (25%), professional services (24%), administrative & support services (24%) and construction (24%). In contrast, employees from community, social & personal services were the least likely to have changed jobs (14%).

2.6 Higher mobility was observed among employees in occupations with higher prevalence of contract and casual/on-call employment. As such, employees previously working in service & sales jobs (23%) saw the highest turnover among the various occupational groups. This was followed by cleaners & labourers (22%) and plant & machine operators & assemblers (21%). The least mobile were clerical workers with only 16 out of 100 workers having changed jobs over the last two years.

[8] $1,500 is the income cut off used in determining eligibility for Workfare Income Supplement for low-wage workers in 2009.
[9] $2,500 is the median income of the employee group under study.

Table 1: Proportion of Resident Employees Who Changed Jobs Over the Last Two Years by Socio-demographic Characteristics

Per Cent

Overall	Proportion
Overall	19.5
Age (years)	
Below 30	28.0
30-34	25.6
35-39	19.9
40-54	16.5
55-60	11.7
Above 60	11.4
Sex	
Male	20.6
Female	18.2
Marital Status	
Single	24.0
Married	17.5
Divorced / Widowed	21.8
Highest Qualification Attained (excluding WSQ)	
Primary & Below	19.1
Lower Seccondary	19.7
Secondary	17.4
Upper Secondary	19.1
Diploma	20.2
Degree	21.3
WSQ	
WSQ	28.1
No WSQ	19.3
Employment Type	
Contract	33.8
Permanent	17.6
Employment Nature	
Part-time (Under-employed)	27.1
Part-time (Not under-employed)	17.3
Full-time	19.2
Income Quartile	
Below $1,500	24.7
$1,500 to $2,499	19.9
$2,500 to $4,332	18.1
$4,333 and above	15.2

Source: Labour Force Survey, MOM.

Table 1: Proportion of Resident Employees Who Changed Jobs Over the Last Two Years by Socio-demographic Characteristics (Cont')

Per Cent

Previous Sector/Industry	Proportion
Manufacturing	18.9
Construction	23.5
Services	19.4
Hotels & Restaurants	*26.7*
Information & Communication	*25.3*
Professional Services	*24.1*
Administrative & Support Services	*23.7*
Financial Services	*20.1*
Transport & Storage	*18.7*
Wholesale & Retail Trade	*18.2*
Real Estate & Leasing Activities	*18.1*
Community, Social & Personal Services	*13.8*
Previous Occupation	
Managers & Administrators	18.1
Professionals	19.6
Associate Professionals & Technicians	19.5
Service & Sales Workers	22.7
Production Craftsmen & Related Workers	17.7
Clerical Workers	16.3
Plant & Machine Operators & Assemblers	21.3
Cleaners, Labourers & Related Workers	21.5

Source: Labour Force Survey, MOM.

Occupational Mobility

2.7 To assess occupational movements, three occupational categories were identified; namely

(i) PMET: **P**rofessionals, **M**anagers, **E**xecutives and **T**echnicians
(ii) SCC: **S**ales & Service workers, **C**lerical workers, Production **C**raftsmen and Related Workers
(iii) OCL: Plant & Machine **O**perators & Assemblers, and **C**leaners, **L**abourers & Related Workers.

2.8 **Among resident employees who switched jobs over the last two years, most moved to occupations with similar skills level**. This horizontal mobility was observed for 90% of PMETs who changed jobs, 75% of SCC and 76% of OCL (Table 2a). Although 8.7% of PMETs who changed jobs transited into SCC occupations and another 1.7% into OCL ones, together they represented only 2.0% of all resident employees in PMET occupations[10] (Table 2b).

2.9 In comparison, proportionately more resident employees from previously OCL type occupations moved to higher skilled based occupational categories: 21% to SCC jobs and 2.8% to PMET jobs. Still, they represented only 5.2% of all resident employees in OCL occupations.

2.10 Likewise, 14% of SCC employees who switched jobs transited to a PMET occupation and 11% moved to an OCL one, collectively representing only 4.6% of all resident employees working in SCC occupations.

Table 2a: Distribution of Resident Employees Who Changed Job(s) Over the Last Two Years

Per Cent

Mobility		Current Occupation		
		PMET	SCC	OCL
Previous Occupation	PMET	89.6	8.7	1.7
	SCC	14.0	75.1	10.9
	OCL	2.8	21.4	75.8

Source: Labour Force Survey, MOM.

Note:
1. Blue shaded cells: Horizontal mobility.

[10] All resident employees in PMET occupations refer to PMET employees who switched job(s) over the last two years and those who did not.

Table 2b: Percentage of Resident Employees Who Changed Job(s) Over the Last Two Years (among all resident employees in the respective category[2])

Per Cent

Mobility		Current Occupation		
		PMET	SCC	OCL
Previous Occupation	PMET	17.2	1.7	0.3
	SCC	2.6	14.2	2.0
	OCL	0.6	4.6	16.3

Source: Labour Force Survey, MOM.

Notes:

1. Blue shaded cells: Horizontal mobility.
2. Include resident employees in that category who switched job(s) over the last two years and those who did not.

Industrial Mobility

2.11 Similar to occupational mobility, **movement by industry was also dominated by employees who moved within industry (intra) than across (inter) industry**. Employees who changed jobs from financial services (61%), construction (60%), community, social & personal services (59%), hotels & restaurants (56%) and information & communications (52%) industries had the highest proportion of intra-industry mobility; while those from real estate & leasing activities (37%) had the lowest (Table 3a). In manufacturing, 47% of resident employees who had changed jobs remained in the sector during a job change while the remaining majority moved into wholesale & retail trade (11%), community, social & personal services (8.5%), transport & storage (6.7%) and administrative & support services (5.9%).

Table 3a: Distribution of Resident Employees Who Changed Job(s) Over the Last Two Years by Industry

Per Cent

Mobility		Current Industry											
		Manu.	Const.	W.Sale & Retail	Tpt & Str	Htl & Rst	ICT	Fin Serv	Real Estate	Prof Serv	Admin & Suppt	Comm & Social	Others[1]
Previous Industry	Manu.	46.5	4.6	11.2	6.7	4.8	3.7	1.7	1.0	4.4	5.9	8.5	1.0
	Const.	8.9	59.8	4.4	5.2	2.6	2.0	1.0	1.6	5.2	3.8	3.8	1.6
	W.Sale & Retail	9.1	3.3	50.2	5.7	6.5	2.6	3.3	1.7	3.0	4.7	9.4	0.4
	Tpt & Str	9.3	5.1	10.1	44.9	4.2	2.0	2.7	1.9	2.5	5.2	10.8	1.3
	Htl & Rst	4.0	2.7	10.6	5.1	55.8	1.6	1.3	1.4	2.1	7.4	7.5	0.5
	ICT	9.1	1.6	6.6	2.4	1.6	51.7	7.2	1.2	6.0	2.2	10.1	0.2
	Fin Serv	3.3	1.0	3.3	2.5	1.9	4.3	61.2	3.1	4.3	2.7	12.2	0.2
	Real Estate	6.3	3.5	10.6	3.5	4.2	2.1	5.6	37.3	3.5	8.5	13.4	1.4
	Prof Serv	10.0	3.0	7.7	3.3	1.0	5.7	6.5	4.5	42.9	2.8	11.8	0.8
	Admin & Suppt	3.0	3.5	6.3	5.6	7.0	2.3	2.3	2.6	5.4	48.4	12.9	0.7
	Comm & Social	5.9	2.8	6.6	3.6	4.9	3.6	4.1	1.4	3.5	4.3	58.7	0.6
	Others[1]	s	s	s	s	s	s	s	s	s	s	s	s

Source: Labour Force Survey, MOM.

Notes:

1. Includes Agriculture, Fishing, Quarrying, Utilities, Sewerage & Waste Management and activities not adequately defined.
2. Blue shaded cells: Intra-industry mobility.
3. s : data is suppressed as the sample size is too small.

2.12 With respect to inter-industry mobility, wholesale & retail trade and community, social & personal services had the highest incidence of inflows from other industries. Specifically, more than 10% of resident employees who had changed jobs from transport & storage, information & communications, financial services, real estate & leasing, professional services and administrative & support services entered community, social & personal services. Likewise, manufacturing (11%), real estate (11%), hotels & restaurants (11%) and transport & storage (10%) had workers who entered wholesale & retail trade.

Table 3b: Percentage of Resident Employees Who Changed Job(s) Over the Last Two Years by Industry (among all resident employees in the respective category[4])

Per Cent

Mobility		Current Industry											
		Manu.	Const.	W.Sale & Retail	Tpt & Str	Htl & Rst	ICT	Fin Serv	Real Estate	Prof Serv	Admin & Suppt	Comm & Social	Others[1]
Previous Industry	Manu.	8.8	0.9	2.1	1.3	0.9	0.7	0.3	0.2	0.8	1.1	1.6	0.2
	Const.	2.1	14.1	1.0	1.2	0.6	0.5	0.2	0.4	1.2	0.9	0.9	0.4
	W.Sale & Retail	1.6	0.6	9.1	1.0	1.2	0.5	0.6	0.3	0.5	0.9	1.7	0.1
	Tpt & Str	1.7	0.9	1.9	8.4	0.8	0.4	0.5	0.3	0.5	1.0	2.0	0.3
	Htl & Rst	1.1	0.7	2.8	1.4	14.9	0.4	0.3	0.4	0.6	2.0	2.0	0.1
	ICT	2.3	0.4	1.7	0.6	0.4	13.1	1.8	0.3	1.5	0.6	2.5	0.1
	Fin Serv	0.7	0.2	0.7	0.5	0.4	0.9	12.3	0.6	0.9	0.5	2.4	-
	Real Estate	1.1	0.6	1.9	0.6	0.8	0.4	1.0	6.8	0.6	1.5	2.4	0.3
	Prof Serv	2.4	0.7	1.8	0.8	0.2	1.4	1.6	1.1	10.3	0.7	2.8	0.2
	Admin & Suppt	0.7	0.8	1.5	1.3	1.7	0.6	0.6	0.6	1.3	11.5	3.0	0.2
	Comm & Social	0.8	0.4	0.9	0.5	0.7	0.5	0.6	0.2	0.5	0.6	8.1	0.1
	Others[1]	2.2	0.9	0.9	0.7	0.4	0.4	0.2	-	1.6	0.9	1.3	3.6

Source: Labour Force Survey, MOM.

Notes:

1. Includes Agriculture, Fishing, Quarrying, Utilities, Sewerage & Waste Management and activities not adequately defined.
2. Blue shaded cells: Intra-industry mobility.
3. ' - ': nil or negligible.
4. Include resident employees in that category who switched job(s) over the last two years and those who did not.

3 Regression Results

3.1 Results from the regression estimation to isolate the impact of various factors influencing mobility are presented for (i) general occupational mobility, (ii) mobility across PMET, SCC and OCL occupations and (iii) intra and inter-industry worker mobility.

Determinants of Mobility: Factors that affect a worker's ability to change jobs

3.2 Due to differences in demographic characteristics, some workers have higher propensities to change jobs than others. Holding other attributes constant, older employees had a lower probability of changing jobs as individuals become more settled in their careers and shoulder increased responsibilities with age (Table 4). Specifically, the odds for those aged 35 and above were all below 1, decreasing progressively to a low of 0.25 for those aged above 60.

3.3 The econometric results confirmed that those who had undergone WSQ (odds: 1.67) were more likely to switch jobs than those without. Better educated employees (upper secondary & above) (odds: at least 1.18) were also more mobile than those with primary & below education.

3.4 In addition, the type and nature of employment influence the probability of changing jobs. Due largely to more flexible employment terms, resident employees on contract were much more likely to change jobs than those on permanent employment. The odds of changing jobs for voluntary part-timers were lower than full-timers, but there was no evidence to suggest that involuntary part-timers had higher odds of changing jobs than full-timers.

3.5 In general, a large incentive for individuals to change jobs is the income earned. The econometric results confirmed that lower income earners were more prone to switching jobs than higher income earners.

3.6 Similar to findings from overseas empirical studies[11], females had a lower probability of changing jobs than males. Being married was not statistically significant in influencing an employee's decision to change jobs. In contrast, being divorced or widowed increased the odds to switch jobs.

3.7 At the industry level, those in hotels & restaurants and construction displayed the highest probability of changing jobs. This was followed by information & communications and those in professional services. Employees in community, social & personal services were the least likely to change jobs.

3.8 Relative to managers & administrators, resident employees in all the other occupational categories exhibited lower probabilities of changing jobs. The decrease observed for higher skilled workers (e.g. professionals, associate professionals) reflects that changing jobs would lead to depreciation and wastage of accumulated occupation-specific knowledge (Jovanovic, 1979). In addition, greater opportunities for learning-by-doing on the

[11] For example, using a panel of skilled young workers in West Germany, Fitzenberger and Kunze (2005) found that occupational mobility was lower for women than men. The report explained that in West Germany, women tend to experience more entry barriers than men.

job reduce the incentives to switch jobs over time (Pissarides, 1994). At the other spectrum, unskilled workers (e.g. cleaners, labourers and related workers) had an even lower probability of changing jobs as the lack of relevant/transferable skills hinder their mobility.

Table 4: Estimated Effects (Odds Ratio) of Personal & Job Characteristics on Labour Mobility of Resident Employees

Variables	Odds Ratio	Std. Error
Age (years) (Reference Group: Below 30)		
30-34	1.07	0.05
35-39	0.79**	0.04
40-54	0.57**	0.03
55-60	0.33**	0.02
Above 60	0.25**	0.02
Sex (Reference Group: Male)		
Female Dummy	0.79**	0.02
Marital Status (Reference Group: Single)		
Married	1.01	0.04
Divorced / Widowed	1.42**	0.10
Highest Qualification Attained (excluding WSQ) (Reference Group: Primary & Below)		
Lower Secondary	1.04	0.06
Secondary	1.04	0.06
Upper Secondary	1.24**	0.09
Diploma	1.18**	0.09
Degree	1.67**	0.13
WSQ Qualifications (Reference Group: No WSQ)		
WSQ	1.67**	0.16
Employment Type (Reference Group: Permanent Employment)		
Contract	2.22**	0.09
Part-time (Involuntary)	0.92	0.07
Part-time (Voluntary)	0.65**	0.06
Income Range (Reference Group: Below $1,500)		
$1,500 to $2,499	0.63**	0.03
$2,500 to $4,332	0.44**	0.02
$4,333 and above	0.32**	0.02

Source: Labour Force Survey, MOM.

Notes:

1. Green: Statistically Significant & increasing Odds.
2. Red: Statistically Significant & decreasing Odds.
3. * Statistically Significant at 10%.
4. ** Statistically Significant at 5%.

Table 4: Estimated Effects (Odds Ratio) of Personal & Job Characteristics on Labour Mobility of Resident Employees (Cont')

Variables	Odds Ratio	Std. Error
Previous Sector/Industry (Reference Group: Manufacturing)		
Construction	1.35**	0.09
Wholesale & Retail Trade	0.91*	0.05
Transport & Storage	0.97	0.06
Hotels & Restaurants	1.37**	0.09
Information & Communications	1.30**	0.08
Financial Services	1.10	0.07
Real Estate & Leasing Activities	0.99	0.10
Professional Services	1.28**	0.08
Administrative & Support Service Activities	1.15**	0.08
Community, Social & Personal Services	0.62**	0.03
Others[1]	0.70**	0.10
Previous Occupation (Reference Group: Managers & Administrators)		
Professionals	0.83**	0.05
Associate Professionals & Technicians	0.86**	0.05
Clerical Workers	0.67**	0.05
Service & Sales Workers	0.84**	0.06
Production Craftsman & Related Workers	0.69**	0.06
Plant & Machine Operators & Assemblers	0.88	0.07
Cleaners, Labourers & Related Workers	0.78**	0.07

Source: Labour Force Survey, MOM.

Notes:

1. Includes Agriculture, Fishing, Quarrying, Utilities, Sewerage & Waste Management and activities not adequately defined.
2. Green: Statistically Significant & increasing Odds.
3. Red: Statistically Significant & decreasing Odds.
4. * Statistically Significant at 10%.
5. ** Statistically Significant at 5%.

Occupational Mobility

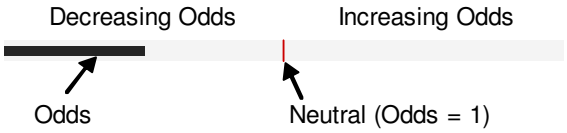
3.9 The base model of labour mobility was extended to investigate whether workers had transited to different occupational groups and the factors that affected the move.

3.10 As a resident employee ages, the odds of transiting to a higher skilled job decreases (Table 5). For example, an OCL employee aged 30-34 had only a 0.49 chance to move up to a SCC job relative to the reference group of employees aged below 30. This likelihood decreased further to 0.15 for those aged above 60. The same was true for resident employees in SCC occupations, with mobility to PMET occupations decreasing from an odd of 0.43 for those aged 40-54 to 0.26 for those aged 55-60 to 0.07 for those aged above 60.

Table 5: Estimated Effects (Odds Ratio) of Age on Occupational Mobility of Resident Employees

Reference group: Age below 30 years

Previous Occupation	Age (years)	Current Occupation: PMET	Current Occupation: SCC	Current Occupation: OCL
PMET	30-34	1.12*	1.09	1.81
	35-39	0.84**	1.36	0.48
	40-54	0.61**	0.89	1.83
	55-60	0.30**	0.42**	1.06
	Above 60	0.35**	0.57	0.12
SCC	30-34	0.98	0.98	1.73*
	35-39	0.71	0.67**	1.16
	40-54	0.43**	0.48**	1.16
	55-60	0.26**	0.28**	0.64
	Above 60	0.07**	0.24**	0.59
OCL	30-34	1.28	0.49**	0.94
	35-39	0.96	0.44**	1.04
	40-54	1.14	0.47**	0.76*
	55-60	0.36	0.22**	0.55**
	Above 60	0.74	0.15**	0.35**



Source: Labour Force Survey, MOM.

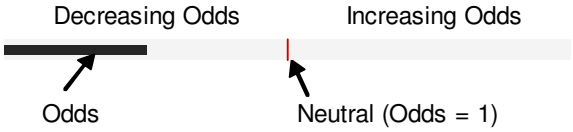
Notes:
1. * Statistically Significant at 10%.
2. ** Statistically Significant at 5%.
3. Blue shaded cells: Horizontal mobility.

3.11 Having **higher formal qualification, especially university education strongly facilitated the move to PMET occupations**. Workers with higher levels of education also exhibited lower probability of switching into OCL type occupations (Table 6).

Table 6: Estimated Effects (Odds Ratio) of Education on Occupational Mobility of Resident Workforce

Reference group: Primary & Below Education

Previous Occupation	Education	Current Occupation: PMET	Current Occupation: SCC	Current Occupation: OCL
PMET	Lower Secondary	1.80	0.94	0.28**
PMET	Secondary	1.49**	0.47*	0.12**
PMET	Upper Secondary	2.52*	0.36**	0.03**
PMET	Diploma	2.33	0.29**	0.01**
PMET	Degree	3.40**	0.16**	0.01**
SCC	Lower Secondary	9.32**	1.16	0.71*
SCC	Secondary	19.87**	1.25**	0.48**
SCC	Upper Secondary	44.28**	1.09	0.32**
SCC	Diploma	74.99**	0.96	0.14**
SCC	Degree	128.66**	1.11	n.a.
OCL	Lower Secondary	7.18*	1.84**	0.83**
OCL	Secondary	21.02**	1.91**	0.83*
OCL	Upper Secondary	83.81**	2.64**	0.76
OCL	Diploma	126.62**	2.24	0.31**
OCL	Degree	331.01**	6.96**	n.a.



Source: Labour Force Survey, MOM.

Notes:
1. * Statistically Significant at 10%.
2. ** Statistically Significant at 5%.
3. Blue shaded cells: Horizontal mobility.
4. n.a.: not available.

3.12 **Similar to the earlier finding that WSQ increased the ease of job mobility, WSQ also significantly increased the chances of OCL employees moving up to a SCC occupation** (odds: 5.07) (Table 7). In addition, WSQ benefited those in PMET and SCC occupations, increasing their odds for lateral mobility across similarly-skilled occupations. However, reflecting the emphasis on more technical and specialised skills within the WSQ training framework, such vocational training was associated with increased chances of employees in previously PMET occupations to move to SCC ones (odds: 2.21). However, there was no statistically significant evidence to suggest that the converse was true.

3.13 Overall, contract workers exhibited high probability of transiting within and across occupational categories. In particular, SCC contract workers were more likely to move to higher skilled PMET jobs. Such transition to a higher skilled occupation was not statistically significant among OCL contract workers.

3.14 Among part time workers, only those on a voluntary basis registered a lower probability within each of the three main occupational classifications. There was no statistical evidence that working part-time impacted the mobility to higher skilled jobs.

Table 7: Estimated Effects (Odds Ratio) of WSQ, Contract Work & Part-time Work on Occupational Mobility of Resident Workforce

Reference group: No WSQ

	WSQ	Current Occupation		
		PMET	SCC	OCL
Previous Occupation	PMET	**1.41****	**2.21****	0.86
	SCC	1.06	**1.71****	1.56
	OCL	2.42	**5.07****	1.06

Reference group: Permanent Workers

	Contract	Current Occupation		
		PMET	SCC	OCL
Previous Occupation	PMET	**2.20****	**2.58****	1.35
	SCC	**1.99****	**2.06****	**1.84****
	OCL	1.19	1.08	**2.07****

Reference group: Full-time Workers

		Part-time	Current Occupation		
			PMET	SCC	OCL
Previous Occupation	PMET	Involuntary	0.95	0.93	2.17
		Voluntary	**0.63****	0.71	1.74
	SCC	Involuntary	0.77	0.88	1.29
		Voluntary	0.75	**0.52****	**1.72****
	OCL	Involuntary	0.82	1.18	0.94
		Voluntary	n.a.	0.95	**0.71****

Source: Labour Force Survey, MOM.

Notes:

1. * Statistically Significant at 10%.
2. ** Statistically Significant at 5%.
3. **Green: Statistically Significant & Increasing Odds.**
4. **Red: Statistically Significant & Decreasing Odds.**
5. Blue shaded cells: Horizontal mobility.
6. n.a.: not available.

Intra and Inter-Industry Mobility

3.15 In this section, the mobility of resident employees within and across three major sectors was investigated; namely (i) manufacturing, (ii) construction and (iii) services.[12]

3.16 The evidence suggested that age reduced the likelihood of intra-industry mobility within the three major sectors (Table 8). For resident employees in services, the likelihood of those aged 30-34 changing jobs within the services industry was 1.12 relative to the reference group of those aged below 30. This likelihood declined progressively to 0.24 for those above 60.

Table 8: Estimated Effects (Odds Ratio) of Age on Intra and Inter-Industry Mobility of Resident Employees

Reference group: Age below 30 years

Previous Industry	Age (years)	Current Industry: Manufacturing	Current Industry: Construction	Current Industry: Services
Manufacturing	30-34	1.23	1.23	0.71**
	35-39	1.06	0.88	0.58**
	40-54	0.83	0.74	0.43**
	55-60	0.46**	0.11**	0.31**
	Above 60	0.41**	0.19	0.23**
Construction	30-34	0.60	1.19	0.79
	35-39	0.58	1.19	0.61
	40-54	0.43	0.90	0.51**
	55-60	0.52	0.57	0.35**
	Above 60	n.a.	0.39**	0.43*
Services	30-34	1.13	1.12	1.12**
	35-39	0.97	0.86	0.77**
	40-54	0.69**	0.72	0.55**
	55-60	0.32**	0.68	0.32**
	Above 60	0.18**	0.47*	0.24**

Decreasing Odds — Increasing Odds

Odds — Neutral (Odds = 1)

Source: Labour Force Survey, MOM.

Notes:
1. * Statistically Significant at 10%.
2. ** Statistically Significant at 5%.
3. Blue shaded cells: Intra-industry mobility.
4. n.a.: not available.

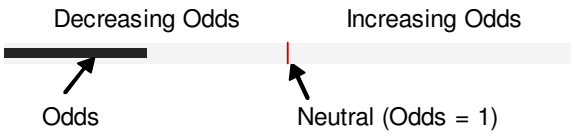
[12] Segmentations by more detailed industries were estimated but not reported as the results were largely statistical insignificant due to the small number of observations in each disaggregated category.

3.17 Formal education, especially **university education increases the probability of intra-industry mobility** for both manufacturing and services industries respectively (Table 9). In terms of switching jobs across sectors, **both upper secondary education and university education enabled greater inter-industry mobility**. In particular, both upper secondary and university education enabled workers to switch jobs from (i) manufacturing to services and from (ii) construction to manufacturing in the case of degree holders, and to services in the case of upper secondary. The relatively low inter-industry mobility conferred by a polytechnic education could be due to its more focused and specialised curriculum.

Table 9: Estimated Effects (Odds Ratio) of Education on Intra and Inter-Industry Mobility of Resident Workforce

Reference group: Primary & Below Education

Previous Industry	Education	Current Industry: Manufacturing	Current Industry: Construction	Current Industry: Services
Manufacturing	Lower Secondary	0.84	1.59	0.90
	Secondary	0.97	1.17	1.16
	Upper Secondary	1.21	1.17	1.88**
	Diploma	1.24	0.77	1.34
	Degree	1.66**	1.24	2.54**
Construction	Lower Secondary	1.18	0.89	1.09
	Secondary	1.62	1.00	1.33
	Upper Secondary	2.53	0.88	2.00*
	Diploma	1.58	0.97	0.93
	Degree	4.08*	1.24	1.52
Services	Lower Secondary	1.38	1.14	1.03
	Secondary	1.19	0.93	0.96
	Upper Secondary	1.08	1.31	1.09
	Diploma	1.55	0.87	1.08
	Degree	1.29	1.21	1.49**



Source: Labour Force Survey, MOM.

Notes:
1. * Statistically Significant at 10%
2. ** Statistically Significant at 5%
3. Blue shaded cells: Intra-industry mobility

3.18 Reflecting the wide spectrum of WSQ programmes for services occupations, resident employees with WSQ were more likely to move within/into services after a job switch than those without WSQ (Table 10). For example, workers who underwent WSQ training would more likely remain within the services sector (odds: 1.64). Similarly, WSQ training also facilitated movement from manufacturing to services (odds: 2.10).

Table 10: Estimated Effects (Odds Ratio) of WSQ on Intra and Inter-Industry Mobility of Resident Workforce

Reference group: No WSQ

WSQ		Current Industry		
		Manufacturing	Construction	Services
Previous Industry	Manufacturing	1.17	0.81	**2.10****
	Construction	n.a.	1.12	2.22
	Services	1.43	0.87	**1.64****

Source: Labour Force Survey, MOM.

Notes:

1. * Statistically Significant at 10%.
2. ** Statistically Significant at 5%.
3. **Green: Statistically Significant & Increasing Odds.**
4. **Red: Statistically Significant & Decreasing Odds.**
5. Blue shaded cells: Intra-industry mobility.
6. n.a.: not available.

4 Conclusion

4.1 Human capital upgrading is required to ensure that workers are mobile and not displaced by economic restructuring or become structurally unemployed due to skills mismatch. This is especially so as older workers generally have lower likelihood of moving into higher skilled jobs.

4.2 The findings of the study showed that education, especially university education enhances movement to PMET occupations. Besides academic qualifications, vocational WSQ training enhances mobility, increasing the chances of plant & machine operators, cleaners and labourers in moving to service, clerical, craftsmen & related occupations. Reflecting the wide spectrum of WSQ programmes for services occupations, WSQ also facilitates the attachment/movement into services after a job switch.

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